

PEDRO HENRIQUE ROCHA DE ANDRADE

Computer Engineering
de Andrade, P. H. R.

Last updated: August 4, 2026

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ACADEMIC SUMMARY

I'm currently a Computer Engineering student with a passion for astronomy and data science. I have experience in scientific research, with a focus on computational astrophysics, and a history of publications in national conferences. I'm always seeking to learn and contribute to the scientific community, exploring new technologies and methodologies to advance our understanding of the universe. I have a hobby in animations and simulations, which allows me to create engaging visualizations to communicate complex concepts in an accessible way, with a focus on science outreach of physics-related content.

ACADEMIC BACKGROUND

Bachelor in Computer Engineering

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

August. 2023 – present Rio de Janeiro, Brazil

Technical Degree in Informatics

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

February. 2020 – December. 2022 Rio de Janeiro, Brazil

RESEARCH PROJECTS

Analysis of Dynamical Simulations of Merging Galaxy Clusters

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

September. 2022 – March. 2023 Rio de Janeiro, Brazil

Junior Scientific Initiation Scholarship - CNPq

Supervisor: Dr. Ana Cecília Soja

Simulating the Impact of Satellites on Astronomical Observations

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

2024 – 2025 Rio de Janeiro, Brazil

Scholarship in the Scientific Initiation project - CNPq

Supervisor: Dr. Ana Cecília Soja

MobFog at IFFMaker

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

August. 2023 – February. 2024 Rio de Janeiro, Brazil

Scholarship in the Scientific Initiation project - IFF

Supervisor: Dr. Ana Cecília Soja

Anomaly Detection in Stars of the Milky Way: Exploring Data from Gaia and Other Surveys with Machine Learning

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

2025 – present Rio de Janeiro, Brazil

Scholarship in the Scientific Initiation project - CNPq

Supervisor: Dr. Ana Cecília Soja and Dr. Maria Luiza Linhares Dantas

EXTERNAL COURSES

Networks

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

August. 2020 Rio de Janeiro, Brazil

Operational Systems

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

August. 2021 Rio de Janeiro, Brazil

Server Administration

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

August. 2022 Rio de Janeiro, Brazil

RESEARCH INTERESTS

EVENTS

Feiras de Ciências

21st Brazilian Science and Engineering Fair

Poster – FEBRACE

2023 • São Paulo, Brazil

40th International Science and Technology Fair

Poster – MOSTRATEC

2023 • Novo Hamburgo, Brazil

IX Brazilian Science Initiation Fair

Poster – FEBIC

2024 • Pomerode, Brazil

XVI Scientific and Technological Exhibition of the Açaí Institute

Poster – MCTIA

2025 • Belém, Brazil

Congressos

44° CONUBES

Participant – CONUBES

2022 • Brasília, Brazil

XV Scientific and Technological Initiation Congress and IX Fluminense Congress of Postgraduate Studies

Oral Presentation – CONFICT

2024 • Campos dos Goytacazes, Brazil

XVI Scientific and Technological Initiation Congress and X Fluminense Congress of Postgraduate Studies

Oral Presentation – CONFICT

2025 • Campos dos Goytacazes, Brazil

XLVIII Annual Meeting of the Brazilian Astronomical Society

Poster – SAB

2025 • Caxambu, Brazil

XVII Scientific and Technological Initiation Congress and XI Fluminense Congress of Postgraduate Studies

Talk – CONFICT

2026 • Campos dos Goytacazes, Brazil

78ª Annual Meeting of Brazilian Society to Science Progress

Poster – SBPC

2026 • Niterói, Brazil

Winter School of Astrophysics

Poster

2026 • Rio de Janeiro, Brazil

XLIX Annual Meeting of the Brazilian Astronomical Society

Poster – SAB

2026 • Natal, Brazil

Teaching, Research and Extension Congress (CONEPE)

Presentation – CONEPE

2026 • Campos dos Goytacazes, Brazil

Semanas Acadêmicas

VIII Knowledge Showcase and I Job Fair

Participant

2020 • Bom Jesus do Itabapoana, Brazil

IX Knowledge Showcase and II Job Fair

Participant

2021 • Bom Jesus do Itabapoana, Brazil

 I Computational Engineering and Management Week

Participant

2021 • Bom Jesus do Itabapoana, Brazil

 X Knowledge Showcase and III Job Fair

Speaker

2022 • Bom Jesus do Itabapoana, Brazil

 II Computational Engineering and Management Week

Participant

2022 • Bom Jesus do Itabapoana, Brazil

 XI Knowledge Showcase and IV Job Fair

Participant

2023 • Bom Jesus do Itabapoana, Brazil

 III Computational Engineering and Management Week

Participant

2023 • Bom Jesus do Itabapoana, Brazil

 XII Knowledge Showcase and V Job Fair

Participant

2024 • Bom Jesus do Itabapoana, Brazil

 IV Computational Engineering and Management Week

Speaker and Organizer

2024 • Bom Jesus do Itabapoana, Brazil

PUBLICATIONS

Link Nasa ADS: [NASA ADS - Pedro H. R. de Andrade](#)

Proceedings

[1] P. H. R. de Andrade, M. L. L. Dantas, and A. C. Soja. « Stellar properties and chemical features of the Gaia Catalogue of Nearby Stars observed by GALAH DR4 ». In: *Boletim da Sociedade Astronômica Brasileira*. Vol. 37. 1. 2026, pp. 308–309.

[2] Pedro Henrique Rocha de Andrade, Ana Cecilia Soja, and Maria Luiza Linhares Dantas. « How varied are stellar populations in the Solar vicinity? Exploring GCNS and GALAH DR4 with t-SNE ». in: *Boletim da Sociedade Astronômica Brasileira*. Natal, RN, 2026.

[3] Pedro Henrique Rocha de Andrade, Ana Cecilia Soja, and Maria Luiza Linhares Dantas. « Propriedades estelares e características químicas do Gaia Catalogue of Nearby Stars observado pelo GALAH DR4 ». In: *Anais do XVI Congresso de Iniciação Científica e Tecnológica e X Congresso Fluminense de Pós-Graduação*. Campos dos Goytacazes, RJ: UENF, 2026.

[4] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Maria Luiza Linhares Dantas. « Detecção de Anomalias em Estrelas da Via Láctea: Explorando Dados do Gaia e Outros Surveys com Aprendizado de Máquina ». In: *Anais da 78ª Reunião Anual da Sociedade Brasileira para o Progresso da Ciência (SBPC)*. Niterói, RJ: UFF, 2026.

[5] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Maria Luiza Linhares Dantas. « Detecção de Anomalias em Estrelas da Via Láctea: Explorando Dados do Gaia e Outros Surveys com Aprendizado de Máquina ». In: *Resumos da Escola de Inverno de Astrofísica do Observatório Nacional*. Rio de Janeiro, RJ: Observatório Nacional, 2026.

[6] Maycon Jorge Deláqua da Silva, Pedro Henrique Rocha de Andrade, and Ana Cecilia Soja. « Simulando o impacto de satélites em observações astronômicas ». In: *Anais do XVI Congresso de Iniciação Científica e Tecnológica e X Congresso Fluminense de Pós-Graduação*. Campos dos Goytacazes, RJ: UENF, 2025.

[7] Maycon Jorge Deláqua da Silva et al. « Simulando o impacto de satélites em observações astronômicas ». In: *XVI Mostra de Ciência e Tecnologia do Instituto Ação (MCTIA)*. 2025.

[8] Maycon Jorge Deláqua da Silva et al. « Simulando o impacto de satélites em observações astronômicas ». In: *Anais do XVI Congresso de Iniciação Científica e Tecnológica e IX Congresso Fluminense de Pós-Graduação*. Campos dos Goytacazes, RJ: UENF, 2024.

[9] Maycon Jorge Deláqua da Silva et al. « Simulando o impacto de satélites em observações astronômicas ». In: *Anais da IX FEBIC – Feira Brasileira de Iniciação Científica*. 2024. ISBN: 978-65-991144-6-5.

[10] Pedro Henrique Rocha de Andrade and Ana Cecília Soja. « Análise de simulações dinâmicas de aglomerados de galáxias em fusão ». In: *Anais da 21ª FEBRACE – Feira Brasileira de Ciências e Engenharia*. Vol. 21. 2023, p. 5. ISBN: 978-65-89190-18-9.

[11] Gabriel Coelho de Souza Poubel et al. « MobFog no IFFMaker ». In: *Anais da XI Mostra do Conhecimento 2023*. 2023, p. 21.

Articles

[12] P. H. R. de Andrade, M. L. L. Dantas, and A. C. Soja. « Stellar properties and chemical features of the Gaia Catalogue of Nearby Stars observed by GALAH DR4 ». In: *arXiv e-prints* (Dec. 2025). DOI: [10.48550/arXiv.2512.10037](#).

AWARDS

- IX FEBIC**
3rd Place in the Undergraduate Category with the work "Simulating the Impact of Satellites on Astronomical Observations"
- XVI MCTIA**
1st Place in the Undergraduate Category with the work "Simulating the Impact of Satellites on Astronomical Observations"

STATISTICS

- 3 Oral Presentations**
National Events
- 7 Posters**
National Events
- 9 Science Fairs/Congresses**
National Events
- 10 Academic Weeks**
Regional Events

PROGRAMMING

Python

Git

Github

LaTeX

C++

Java

NodeJS

LANGUAGES

Portuguese

English

French

Spanish